Happy Pipes

An Outdoor Plumbing IOT Solution to Prevent Pipes Freezing During Sub-Zero Temperatures.

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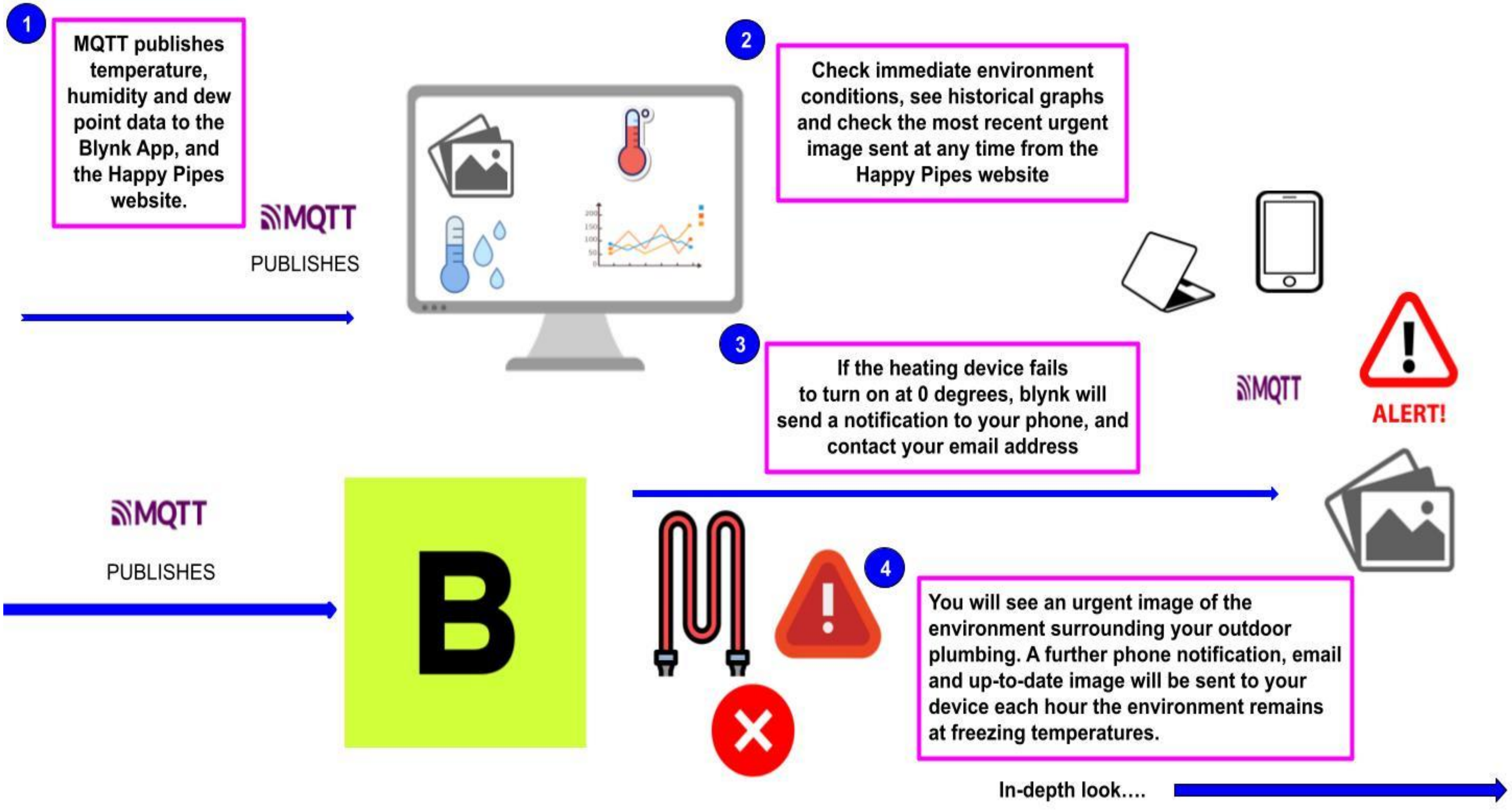
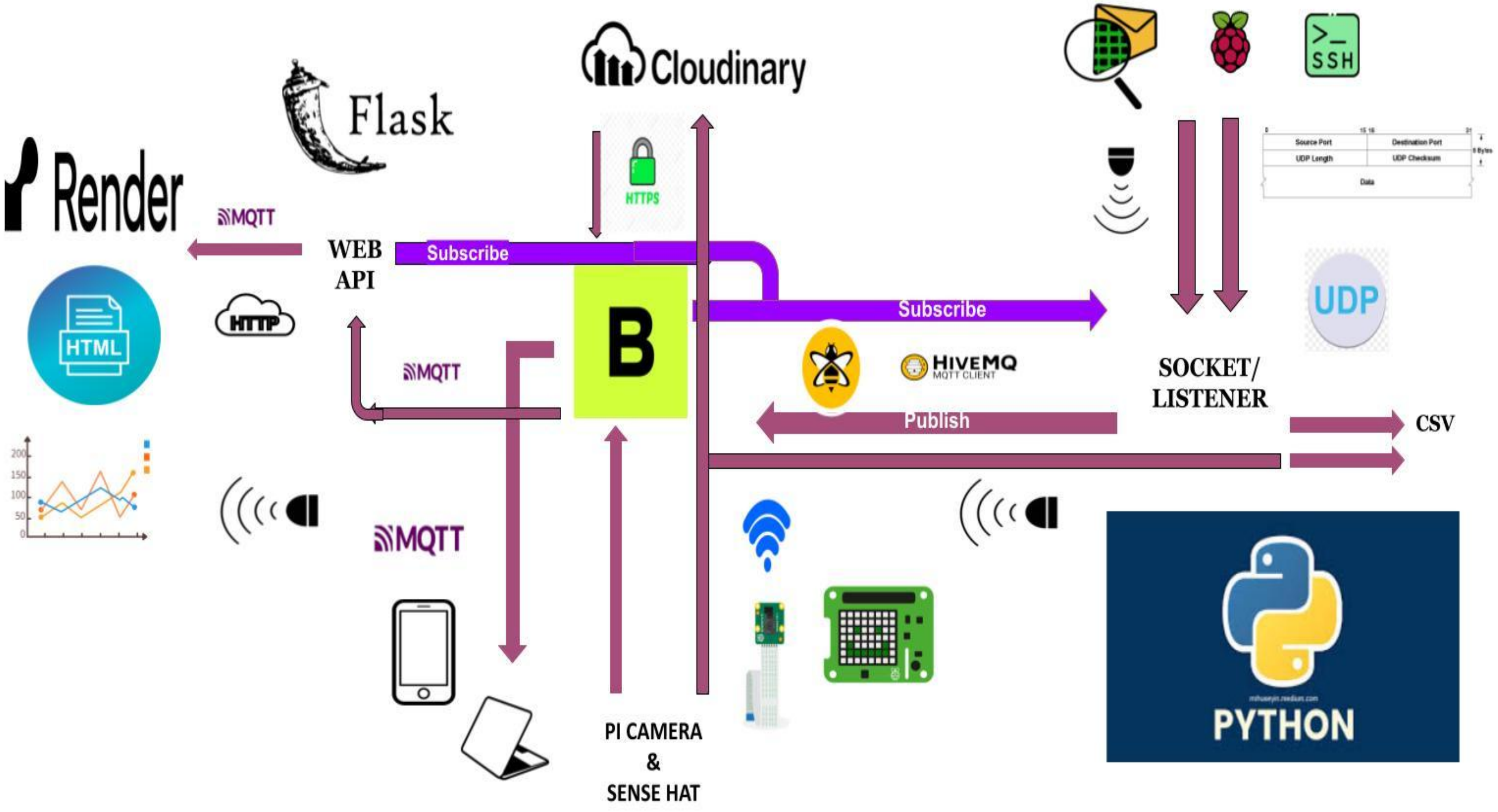

1
MQTT publishes temperature, humidity and dew point data to the Blynk App, and the Happy Pipes website.
MQTT
PUBLISHES
2
Check immediate environment conditions, see historical graphs and check the most recent urgent image sent at any time from the Happy Pipes website
3
If the heating device fails to turn on at 0 degrees, blynk will send a notification to your phone, and contact your email address
MQTT
ALERT!
MQTT
PUBLISHES
B
4
You will see an urgent image of the environment surrounding your outdoor plumbing. A further phone notification, email and up-to-date image will be sent to your device each hour the environment remains at freezing temperatures.
In-depth look....


Render
Flask
Cloudinary
HTTPS
MQTT
WEB
API
Subscribe
HTTP
HTML
MQTT
B
Subscribe
HIVEMQ
MQTT CLIENT
Publish
SOCKET/
LISTENER
UDP
Source Port
Destination Port
UDP Length
UDP Checksum
Data
CSV
MQTT
PI CAMERA
&
SENSE HAT
PYTHON

Happy Pipes - Project Description

This project is a prototype system for monitoring environmental conditions. It demonstrates how sensor data can be collected and used to alert users to potential problems. The system acts as a **fail-safe** in case the outdoor heating element fails, helping to prevent burst pipes or mold formation. The Raspberry Pi simulates readings and monitors both **temperature** and **dew point**. Low temperatures below 0°C could cause burst pipes, while high dew point levels can lead to mold forming on pipes.

Sensor data is sent to the **Blynk app** on your phone, where users receive an **instant phone notification/alert** if conditions become unsafe. If the temperature drops below 0°C, the system triggers the Blynk alert and immediately sends a picture. If the temperature remains below zero, it continues to send an updated image and alert every hour until the environment is safe again (5°C or above).

The Raspberry Pi itself provides a quick visual indicator: the onboard LED is **green** when conditions are safe and **red** when unsafe. All data is also recorded and displayed on the **Happy Pipes website**, allowing users to review historical trends and monitor how conditions evolve over time. Although the system currently uses simulated data, it effectively demonstrates the workflow of a real-world fail-safe monitoring system: sensing, alerting, and historical data visualization.